DVT MSE Question Paper

1. **Calculate Tasks:**
   Mean, Median, Mode
   Variance & Std Dev
   Correlation (Year vs Sales)
2. Difference between gross and net profit
3. Growth rate in net profit
4. Intercept
5. Correlation coefficient of sales and net profit
   Regression and predict for 2035
6. And write the conclusion

Year	Sales	Gross profit	Net profit
2010	5000	400	180
2011	5500	460	210
2012	6000	520	240
2013	6500	490	225
2014	7000	550	270
2015	7500	620	310
2016	8000	700	360
2017	8500	780	410
2018	9000	870	480
2019	9500	980	560
2020	10000	600	400
2021	10500	750	530
2022	11000	680	490
2023	11500	690	510
2024	12000	900	400

2025	12500	1000	680
2026	13000	540	470
2027	13500	980	340
2028	14000	1050	490
2029	14500	1200	600

2. Create data sheets and calculations as described below. Calculate:
Line chart, bar chart and
CGPA
% improvement
Std deviation
Trend prediction
Plot the line chart for semester vs % growth.
What SGPA is needed in next semester to reach CGPA = 9?"
Correlation Analysis Between Semester & SGPA
Best and least SGPA

Semester	SGPA
1	7.1
2	7.5
3	8.0
4	8.2
5	8.5
6	8.7
7	9.0
8	9.2
9	9.3
10	9.5

3. **For the below given table perform Tasks:**

- Mean, Median, Mode of salary experience and age
- Min, Max
- Plot bar and line chart position vs salary and employee experience vs salary and age vs salary

correlation between salary vs age and salary vs experience

Display people who are having salary > 90,000

Age	Salary	Position level	Experience
45	99031	2	8
34	137141	1	6
56	142330	2	18
23	75721	2	5
22	82318	3	9
31	140846	4	7
32	86310	3	19
42	115539	4	14
43	119724	2	3
44	66065	2	5
45	142532	5	12
29	140795	3	19
28	142436	4	14
44	92064	2	20
55	136212	5	12
22	147044	2	13
23	78232	4	7
34	146646	2	4
45	82369	1	1

56	65088	5	7
54	130078	5	14
43	112231	3	9
32	92082	3	10
21	139487	5	2
49	80843	4	4
36	100920	3	12
38	65726	4	2
39	134341	3	15
46	92346	4	4
39	149005	1	19

4a. For the given below chart calculate:

- Pie chart
- % contribution
- Total expense
- Highest & Lowest

Category	Amount
Food	8000
Travel	3000
EMI	15000
Bills	6000
Entertainment	2500
Savings	5000

- Bar chart
- Average sales

Product	Sales
P1	150
P2	250
P3	350
P4	450
P5	550
P6	650
P7	750
P8	850
P9	950
P10	1050
P11	1150
P12	1250
P13	1350
P14	1450
P15	1550

4b. Create data sheet for Age of 100 persons (use random function to get the age of a person.

5. consider the employees data set and do the following tasks using python

- Display first and last few rows of the data
- Display selected columns Position and salary
- Display the rows with salary >9000
- Create a column TDS with TDS=0.10*Salary

- display only the web developer data
- plot the scatter plot of salary vs experience.
- sort the data based on years of experience
- perform the corelation between age and salary and experience vs salary
- display only male and only female employees

First Name	Gender	Age	Job Title	Years Of Experience	Salary
Jose	male	25	Project Manager	1	8500
Diane	female	26	Machine Learning Engineer	2	7000
Shawn	male	37	Project Manager	14	17000
Brenda	female	31	Web Developer	8	10000
Sean	male	35	Project Manager	11	14500
Joshua	male	28	Project Manager	3	10500
Brianna	female	33	Machine Learning Engineer	10	11000
John	male	33	Mobile Developer	10	11000
Jillian	female	34	Web Developer	10	11000
Melanie	female	41	Web Developer	15	14500
Brandy	female	37	Web Developer	14	14000
Robert	male	36	DevOps Engineer	12	15000
George	male	28	Machine Learning Engineer	4	10000
Joseph	male	40	Machine Learning Engineer	14	14000
Dana	female	32	Project Manager	8	12000
Christopher	male	29	Web Developer	5	7500
Nicole	female	26	Designer	4	10000
Peter	male	28	Mobile Developer	3	6500
Sergio	male	31	Tester	6	9000
Brianna	female	31	Designer	5	10500
Taylor	female	32	Machine Learning Engineer	6	11000
Michael	male	29	Tester	5	9500
Yvette	female	26	Designer	2	7000
Kristina	female	34	DevOps Engineer	11	12500
Vanessa	female	36	Machine Learning Engineer	10	13000
Robert	male	33	Tester	9	9500
Tammy	female	37	HR Manager	14	14000
Rodney	male	28	DevOps Engineer	3	9500
David	male	25	Designer	2	7000
Brian	male	31	Machine Learning Engineer	8	10000

6.For the given data set perform the below task Tasks:

- Mean, Median, Variance
- Quartiles
- Histogram
- Outlier detection

Age
5
7
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7.For the given table perform Tasks:

- Display rows
- Filter salary >10000
- Add TDS column
- Scatter plot (experience vs salary)
- Sort data

Calculate the correlation between experience and salary

Name	Experience	Salary
Emp1	1	6000
Emp2	2	7000
Emp3	3	8000
Emp4	4	9000

Emp5	5	10000
Emp6	6	11000
Emp7	7	12000
Emp8	8	13000
Emp9	9	14000
Emp10	10	15000
Emp11	11	16000
Emp12	12	17000
Emp13	13	18000
Emp14	14	19000
Emp15	15	20000
Emp16	16	21000
Emp17	17	22000
Emp18	18	23000
Emp19	19	24000
Emp20	20	25000

8. Consider the given data set and calculate the mean and average std variation and correlation of the covid cases and also calculate the growth rate in each day and plot the graph of day vs growth rate.

Day	Cases
1	100
2	150
3	200
4	300
5	500

6	400
7	450
8	600
9	700
10	800
11	750
12	900
13	1000
14	950
15	1100
16	1200
17	1300
18	1250
19	1400
20	1500

9. consider covid dataset and create

a.Global COVID-19 Tracker Dashboard: - Create a dashboard to track global COVID-19 cases, deaths, and recoveries

- Map to show geographic distribution of cases
- Bar chart to show top 10 countries with most cases
- Line chart to show daily new cases and deaths

b.COVID-19 Vaccination Progress Dashboard Map to show

- geographic distribution of vaccination rates
- Bar chart to show top 10 countries with highest vaccination rates
- Line chart to show daily new vaccinations

c. Covid severity vs health infrastructure dashboard

- map to show the death rate in countries
- bar chart to show top 10 countries with lowest hospital beds

- line chart to show the diabities prevalence and total death per million

10. consider covid dataset and create

COVID-19 Hospitalization and ICU Dashboard

- Map to show geographic distribution of hospitalization rates
- Bar chart to show top 10 countries with highest hospitalization rates
- Line chart to show daily new hospitalizations and ICU admissions

COVID-19 Testing and Contact Tracing Dashboard

- Bar chart to show top 10 countries with highest testing rates
- Line chart to show daily new tests and contact tracing efforts
- Map to show geographic distribution of testing and contact tracing efforts

c. Covid severity vs health infrastructure dashboard

- map to show the death rate in countries
- bar chart to show top 10 countries with lowest hospital beds
- line chart to show the diabities prevalence and total death per million